PAVEMENT ENGINEERING TECHNICAL CALCULATION NOTE

DOC REF: PAVE-CALC-9516

Mechanistic-Empirical Continuum Analysis • IRC:37-2018 & AASHTO Framework

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Model Surrogate: Multi-Output Random Forest Regressor (v2.4-ML)

RELIABILITY: 90% (Class-I Highway)

DESIGN VERIFICATION STATUS: DESIGN COMPLIANT (PASS)

Standard Traffic: 50.0 MSA | Corridor Area: 7,000 m²

1. STRUCTURAL LAYER ARCHITECTURE & DESIGN INPUTS

Component	Dimension / Value	Mechanical Properties	Specification Notes
Design Traffic Loading	50.0 MSA	Axle Load: 80 kN Dual Tire (0.56 MPa Contact)	IRC:37-2018 Standard Spectrum
Bituminous Layer (Top Course)	150 mm	Elastic Modulus E• = 3,00	Wearing Course (BC) + DBM Course
Granular Layer (Base & Subbase)	250 mm	Elastic Modulus E, = 250	Crushed Stone (WMM) + Granular Subbase (GSB)
Compacted Soil Subgrade	Semi-Infinite Foundation	Resilient Modulus Ef = 50	Estimated CBR "H 5.0% (Native Sub
Road Stretch Geometry	Length: 1.0 km	Width: 7.0 m	Carriageway Footprint: 7,000 m²

2. MECHANISTIC STRAIN RESPONSES & FAILURE CRITERIA

Mechanic Response Parameter	Surrogate Value	Allowable Threshold	Safety Margin / Life	Cumulative Damage	Verification
Tensile Strain (TS) -	178.4 µµ	"d 240.0 µµ	+25.7% (Margin)	Damage DF = 0.59	**PASS (Optimal)**
Compressive Strain (	442.8 µµ	"d 620.0 µµ	+28.6% (Margin)	Damage DR = 0.45	**PASS (Safe)**
Fatigue Criteria Coefficient	1.34e+41	1.00e+35 Minimum	Nf = 84.20 Million Axles	IRC:37-2018 Eq 3.1	**VERIFIED**
Rutting Criteria Coefficient	1.70e-16	1.00e-14 Upper Limit	Nr = 112.00 Million Axles	AASHTO M-E Rutting Eq	**VERIFIED**

3. MATERIAL QUANTITIES & LOGISTICAL ESTIMATION (BoQ)

Material Layer Description	Volumetric Formula	Volume (m^3)	Compacted Density	Net Mass (MT)	Logistics (20 MT Trucks)
Bituminous Courses (BC + DBM)	L (km) × 1000 × W 1000)	1,050.00 m^3	2.40 t/m^3	2,520.00 MT	"H 126 Dumpers (20 MT)
Granular Base & Subbase (WMM + GSB)	L (km) × 1000 × W 1000)	1,750.00 m^3	2.10 t/m^3	3,675.00 MT	"H 184 Dumpers (20 MT)
TOTAL STRUCTURAL PAVEMENT	Total Thickness = 400 mm	2,800.00 m^3	Composite	6,195.00 MT	"H 310 Total Trips

ENGINEERING CERTIFICATION & SIGN-OFF

I hereby certify that the mechanistic responses and material estimations presented herein have been validated against IRC:37-2018 standards utilizing the verified Multi-Output Random Forest surrogate continuum formulation.

IRC:37 VERIFIED

Prepared By: Pavement Design Engineer

Approved By: Lead Highway Consultant